

LAB NO: 08

LAB TITLE : Two-Dimensional Arrays in C++

OBJECTIVE :

The objective of this lab is to understand the concept of two-dimensional arrays in C++. This lab helps students learn how to declare and initialize two-dimensional arrays, access array elements using rows and columns, use nested loops with arrays, calculate row and column sums, and find the maximum value from a matrix. Two-dimensional arrays are useful for storing data in tabular form such as rows and columns.

THEORY :

A two-dimensional array is a collection of one-dimensional arrays. It is used to store data in the form of rows and columns like a table or matrix. Every element in a two-dimensional array is identified by two indexes: row index and column index.

In C++, indexing starts from 0.

For example

```
int matrix[3][3];
```

Array Declaration :

Syntax :

```
dataType arrayName[rows][columns];
```

Array Initialization :

Syntax:

```
int matrix[2][2] = {{1,2},{3,4}};
```

Accessing Array Elements :

Syntax :

```
arrayName[row][column];
```

Nested Loop In 2-D :

Nested loops are used with two-dimensional arrays because one loop handles rows while the second loop handles columns.

SOFTWARE

1. DEV-C++ IDE
2. C++ Compiler

LAB TASKS

Two-Dimensional Array Declaration and Initialization

1. Declare a 2D integer array called "matrix" with dimensions 3x3.
2. Initialize the array with values 1, 2, 3, 4, 5, 6, 7, 8, and 9
3. Display the values of the array elements using nested loops.

```
#include <iostream>
using namespace std;

int main()
{
    int matrix[3][3] =
    {
        {1, 2, 3},
        {4, 5, 6},
        {7, 8, 9}
    }; // matrix ko initialize karna

    for (int i = 0; i < 3; i++)
    {
        for (int j = 0; j < 3; j++)
        {
            cout << matrix[i][j] << " "; // matrix ke elements print karna
        }
    }
}
```

```

    }

    cout << endl;
}

return 0;
}

```

Sum of Rows and Columns in a Two-Dimensional Array

- Declare and initialize a 3x3 integer array with random values.
- Calculate and display the sum of each row and each column separately.
- Display the total sum of all the elements in the array.

```

#include <iostream>
using namespace std;

int main()
{
    int matrix[3][3] =
    {
        {1, 2, 3},
        {4, 5, 6},
        {7, 8, 9}
    };

    int total = 0;

    // Row sums calculate karna
    for (int i = 0; i < 3; i++)
    {
        int rowSum = 0;

        for (int j = 0; j < 3; j++)
        {
            rowSum = rowSum + matrix[i][j];
        }
    }
}

```

```

        cout << "Row " << i + 1 << " Sum = " << rowSum << endl;
    }

    cout << endl;

    // Column sums calculate karna
    for (int j = 0; j < 3; j++)
    {
        int colSum = 0;

        for (int i = 0; i < 3; i++)
        {
            colSum = colSum + matrix[i][j];
        }

        cout << "Column " << j + 1 << " Sum = " << colSum << endl;
    }

    for (int i = 0; i < 3; i++)
    {
        for (int j = 0; j < 3; j++)
        {
            total = total + matrix[i][j]; // Total sum calculate karna
        }
    }

    cout << endl;
    cout << "Total Sum = " << total;

    return 0;
}

```

Finding the Maximum Value in a Two-Dimensional Array • Declare and initialize a 4x4 integer array with random values • Find and display the maximum value in the array.

```

#include <iostream>
using namespace std;

int main()
{
    int matrix[4][4] =
    {
        {12, 7, 9, 15},
        {3, 25, 11, 8},
        {14, 6, 30, 10},
        {2, 18, 5, 20}
    };

    int max = matrix[0][0]; // pehli value ko maximum manna

    for (int i = 0; i < 4; i++)
    {
        for (int j = 0; j < 4; j++)
        {
            if (matrix[i][j] > max)
            {
                max = matrix[i][j]; // maximum value update karna
            }
        }
    }

    cout << "Maximum Value = " << max;

    return 0;
}

```

CONCLUSION

In this lab, we learned the concepts of two-dimensional arrays in C++. We studied declaration, initialization, accessing elements, nested loops, row and column sums, and finding maximum values

from arrays. This lab improved our understanding of matrices and two-dimensional data handling in programming.